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VPC Endpoints



yakubu abdullahi yusuf

vpce-0f14f3488f83d234c / NextWork VPC Endpoint			
Details			
Endpoint ID vpce-0f14f3488f83d234c	Status Available	Creation time Friday, 16 January 2026 at 02:55:44 GMT+3	Endpoint type Gateway
VPC ID vpc-0e9cb8865185b629b (NextWorkS3-vpc)	Status message -	Service name com.amazonaws.us-east-2.s3	Private DNS names enabled No
DNS record IP type service-defined	IP address type ipv4	Service region us-east-2	Private DNS preference -
Private DNS specified domains			



yakubu abdullahi yusuf

NextWork Student

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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is... and it is useful because Amazon VPC is a service that lets you create your own private network inside AWS where you can run cloud resources like EC2 instances. It is useful because it gives you full control over networking and security.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a VPC, launch an EC2 instance in the VPC and connect to an S3 bucket through EC2 and also uploaded a file through AWS CLI to the S3 bucket and also created an endpoint for the secure and private access of AWS services.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how fun it is to use AWS CLI to access different AWS services privately.



yakubu abdullahi yusuf

NextWork Student

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This project took me...

This project took me 30 minutes.



yakubu abdullahi yusuf

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In the first part of my project...

Step 1 - Architecture set up

In this step, I will create a VPC, launch an EC2 instance, connect to the EC2 instance through EC2 instance connect and then set up an S3 bucket.

Step 2 - Connect to EC2 instance

In this step, I will connect to my EC2 instance through EC2 instance connect.

Step 3 - Set up access keys

In this step, I will give my EC2 instance access to AWS services through Access keys.

Step 4 - Interact with S3 bucket

In this step, I will be interacting with my S3 bucket through my EC2 instance.



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Architecture set up

I started my project by launching an EC2 instance linking it to My VPC.

I also set up an S3 bucket and uploaded some files in it.

Summary

Destination

s3://nextwork-vpc-project-sthree

Succeeded

2 files, 4.6 MB (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (2 total, 4.6 MB)

Find by name

Name	Folder	Type	Size	Status	Error
NextWork - Denzel is awesome...	-	image/png	2.3 MB	Succeeded	-
NextWork - Lelo is awesome.pn...	-	image/png	2.3 MB	Succeeded	-



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Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, I configured my AWS CLI with AWS Access Key ID and AWS Secret Access Key, region name and output format.

Access keys are a set of credentials that allow access to AWS services via the CLI.

Secret access keys are like the password that pairs with your access key ID to access AWS services

Best practice

Although I'm using access keys in this project, a best practice alternative is to use AWS CloudShell, a browser-based CLI, to run commands. Use the AWS CLI V2 to enable authentication through a user in IAM identity center.



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Connecting to my S3 bucket

The command I ran was `aws s3 ls`. This command is used to list all the S3 an account.

The terminal responded withwith the name of my s3 bucket. This indicated the access key configuration worked and s3 is available, and I can access AWS services through AWS CLI.

```
default output format [None]: [ec2-user@ip-10-0-9-194 ~]$ aws s3 ls
026-01-15 22:49:05 nextwork-vpc-project-sthree
ec2-user@ip-10-0-9-194 ~]$
```



Connecting to my S3 bucket

I also tested the command `aws s3 ls s3://nextwork-vpc-project-sthree` which returned the objects in my S3 bucket.

```
Default output format [None]: [ec2-user@ip-10-0-9-194 ~]$ aws s3 ls
2026-01-15 22:49:05 nextwork-vpc-project-sthree
[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls s3://nextwork-vpc-project-sthree
2026-01-15 22:51:14      2431554 NextWork - Denzel is awesome.png
2026-01-15 22:51:15      2399812 NextWork - Lelo is awesome.png
[ec2-user@ip-10-0-9-194 ~]$
```



Uploading objects to S3

To upload a new file to my bucket, I first ran the command "sudo touch/tmp/test.txt". This command creates a new file.

The second command I ran was `aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-sthree`. This command will copy the file test.txt into my S3 bucket.

The third command I ran was `aws s3 ls s3://nextwork-vpc-project-sthree`, which validated that the file was successfully copied to the S3 bucket.

```
[ec2-user@ip-10-0-9-194 ~]$ sudo touch /tmp/test.txt
[ec2-user@ip-10-0-9-194 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-sthree
upload: ../../tmp/test.txt to s3://nextwork-vpc-project-sthree/test.txt
[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls
2026-01-15 22:49:05 nextwork-vpc-project-sthree
[ec2-user@ip-10-0-9-194 ~]$ aws s3 ls s3://nextwork-vpc-project-sthree
2026-01-15 22:51:14      2431554 NextWork - Denzel is awesome.png
2026-01-15 22:51:15      2399812 NextWork - Lelo is awesome.png
2026-01-15 23:11:24           0 test.txt
[ec2-user@ip-10-0-9-194 ~]$
```



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In the second part of my project...

Step 5 - Set up a Gateway

In this step, I will be creating an endpoint to ensure a seccure connection within AWS services.

Step 6 - Bucket policies

I just created an S3 endpoint, How do I know it works? We would have to check that out together by limiting my S3 bucket's traffic from my endpoint.

Step 7 - Update route tables

HEre I will be testing my VPC endpoint and I might run into a tiny issue that I have to resolve.

Step 8 - Validate endpoint conection

In this step, I will try to access my S3 bucket again and test my VPC endpoint.



Setting up a Gateway

I set up an S3 Gateway, which is a type of endpoint used specifically for Amazon S3.

What are endpoints?

An endpoint is a service that allows private connections between your VPC and other AWS services without needing the traffic to go over the internet.

vpce-0f14f3488f83d234c / NextWork VPC Endpoint			
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Bucket policies

A bucket policy is a type of IAM policy designed for setting access permissions to an S3 bucket. Using bucket policies, you get to decide who can access the bucket and what actions they can perform with it.

My bucket policy will deny all S3 actions for requests not coming from my VPC endpoint, effectively restricting access to private VPC traffic.

Policy

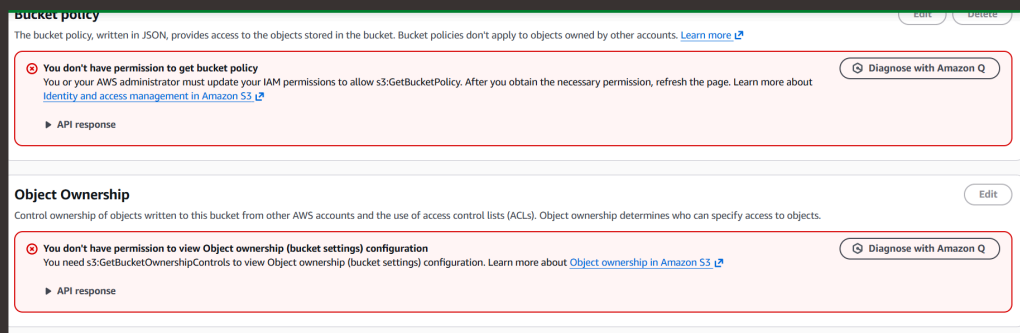
```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Deny",
6       "Principal": "*",
7       "Action": "s3:*",
8       "Resource": [
9         "arn:aws:s3:::nextwork-vpc-project-sthree",
10        "arn:aws:s3:::nextwork-vpc-project-sthree/*"
11      ],
12      "Condition": {
13        "StringNotEquals": {
14          "aws:sourceVpc": "vpce-0f14f3488f83d234c"
15        }
16      }
17    }
18  ]
19 }
20
```




Bucket policies

Right after saving my bucket policy, my S3 bucket page showed 'denied access' warnings. This was because the policy denies all actions unless they come from your VPC endpoint. This means any attempt to access your bucket from other sources, including the AWS Management Console, is blocked!

I also had to update my route table because there isn't a route which enables connection from my EC2 instance to S3 bucket.

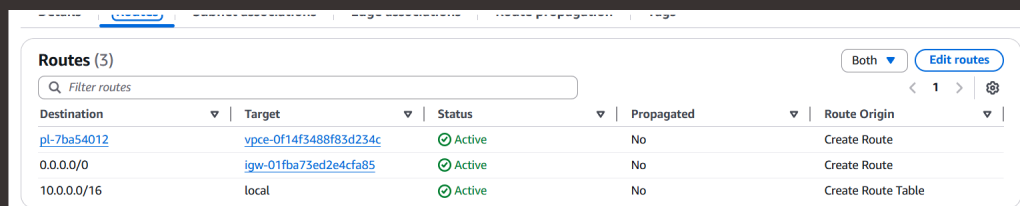




Route table updates

To update my route table, I added my public route table to my endpoint.

After updating my public subnet's route table, my terminal could return the lists of object in my S3 bucket.



The screenshot shows the AWS Management Console 'Routes' page for a specific VPC. It displays a table with three routes. The first route is for destination 'pl-7ba54012' with target 'vpce-0f14f3488f63d234c'. The second route is for destination '0.0.0.0/0' with target 'igw-01fba73ed2e4cfa85'. The third route is for destination '10.0.0.0/16' with target 'local'. All routes are 'Active' and 'Propagated' is 'No'.

Destination	Target	Status	Propagated	Route Origin
pl-7ba54012	vpce-0f14f3488f63d234c	Active	No	Create Route
0.0.0.0/0	igw-01fba73ed2e4cfa85	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table



Endpoint policies

An endpoint policy defines the permissions for traffic going through a VPC endpoint, effectively controlling which services or resources the endpoint can access.

I updated my endpoint's policy by changing the Effect to Deny, I could see the effect of this right away, because i couldnt access the objects in my S3 bucket.

```
Policy
VPC endpoint policy controls access to the service
1 {
2   "Version": "2008-10-17",
3   "Statement": [
4     {
5       "Effect": "Deny",
6       "Principal": "*",
7       "Action": "*",
8       "Resource": "*"
9     }
10  ]
11 }
```



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